

CCNA - Routing and switching

Day 1: Static Routing Concepts Theory :What is static routing?Advantages and disadvantages of static routes. Practical :Configure static routes between two routers in Cisco Packet Tracer

Task 1 :- Open Cisco Packet Tracer and set up a simple network with two routers connected via a serial cable, assigning IP addresses to both router interfaces.

ANS :-

1. Add HWIC-2T Serial Port (Both Routers)

Click Router → Physical tab → turn Power OFF → drag HWIC-2T module into an empty slot → turn Power ON.

2. Connect Cable

Select Connections (Lightning icon) → choose Serial DCE cable → click Router 1 (Serial0/3/0) → click Router 2 (Serial0/3/0).

3. IP Configuration

Router 1 (R1)

Click R1 → CLI tab → press Enter:

```
enable
configure terminal
interface Serial0/3/0
ip address 10.0.0.1 255.255.255.0
clock rate 64000
no shutdown
```

Router 2 (R2) :-

Click R2 → CLI tab → press Enter:

```
enable
configure terminal
interface Serial0/3/0
ip address 10.0.0.2 255.255.255.0
no shutdown
```

Check : On R1, open CLI and type ping 10.0.0.2. If you see Success rate is 80 or 100 percent, the link is working!

Task 2 :- On Router1 in Cisco Packet Tracer, configure a static route so it can reach a PC connected to Router2's LAN network. Test connectivity using the ping command from Router1 to the PC.

ANS :-

Assuming Router1 is connected to Router2, Router2's LAN is 192.168.20.0/24, the PC IP is 192.168.20.2, and Router2's serial IP (next hop) is 10.0.0.2 :

1. Add Static Route on Router1

Click Router1 → CLI tab → run:

```
enable
configure terminal
ip route 192.168.20.0 255.255.255.0 10.0.0.2
end
```

2. Test Ping

On Router1 CLI, run :

```
ping 192.168.20.2
```

Verification Check :

You should see Success rate is 80 or 100 percent (the first packet may drop due to ARP, which is normal).

Task 3 :- Simulate a Zomato-style delivery network : imagine Router1 represents an Ahmedabad branch and Router2 represents a Surat branch. Configure static routes so that devices in each city's LAN can communicate with each other through the routers.

ANS :-

Network IP Plan

- Ahmedabad (Router 1) : LAN = 192.168.10.0/24 | Serial IP = 10.0.0.1
- Surat (Router 2) : LAN = 192.168.20.0/24 | Serial IP = 10.0.0.2

1. Ahmedabad (Router 1) Setup :-

Tells Ahmedabad how to reach Surat's LAN :

Click Router 1 → CLI tab → run :

```
enable
configure terminal
ip route 192.168.20.0 255.255.255.0 10.0.0.2
end
```

2. Surat (Router 2) Setup :-

Tells Surat how to reach Ahmedabad's LAN :

Click Router 2 → CLI tab → run :

```
enable
configure terminal
ip route 192.168.10.0 255.255.255.0 10.0.0.1
end
```

3. Test Connectivity :-

Open Command Prompt on an Ahmedabad PC (192.168.10.2) and ping a Surat PC (192.168.20.2) :

```
ping 192.168.20.2
```

Verification Check :

Seeing Reply from 192.168.20.2: bytes=32... means the two city branches are connected!

Task 4 :- List two advantages and two disadvantages of static routing compared to dynamic routing, using examples from real-life apps like IRCTC or Swiggy to illustrate your points.

ANS :-

Advantages :-

1. Better Security & Control :-

- Why: You choose the exact path—no unauthorized rerouting.
- Swiggy Example: Swiggy sends payment data through a fixed, direct path to HDFC bank so no untrusted network touches credit card details.

2. Saves CPU & Bandwidth :-

- Why: Routers don't waste power sending constant route updates.
- IRCTC Example: IRCTC's small notification server (SMS/Email alerts) uses a fixed static route to save server capacity for heavy Tatkal ticket traffic.

Disadvantages :-

1. No Auto-Reroute on Link Failure :-

- Why: If a cable breaks, traffic stops until an admin fixes it manually.
- Swiggy Example: If a primary network line cuts during dinner peak, Swiggy food orders freeze and time out instead of automatically switching to a backup line.

2. Hard to Manage in Large Networks

- Why: You must manually type routes into every single router when adding new branch networks.
- IRCTC Example: If IRCTC opens 50 new station kiosks across India, using static routing means an engineer must manually program hundreds of routers line by line.